

Homework 27

The Game of Nim

Due by noon on November 17 (Monday)

This is project 6.2 on page 285 in your textbook (I've reproduced all of the material from the text, here, though, so you don't have to look it up.) This is a larger, "capstone" problem from Chapter 6.

The game of Nim is a well-known game with a number of variants. For this project, you'll build the following variant, which has an interesting winning strategy. Two players alternately take marbles from a pile. In each move, a player chooses how many marbles to take. The player must take at least one, but at most half of the marbles. Then, the other player takes a turn. The player who takes the last marble loses.

Here's one sample run to give you an example of how the program should act:

```
Welcome to the game of Nim. Pile Size: 67 marbles

Your move. How many would you like to take? 5

Current pile size: 62 marbles. I'll take: 21
Current pile size: 41 marbles. How many would you like to take? 11
Current pile size: 30 marbles. I'll take: 2
Current pile size: 28 marbles. How many would you like to take? 8
Current pile size: 20 marbles. I'll take: 9
Current pile size: 11 marbles. How many would you like to take? 8
Illegal Move. Maximum you may take is 5 marbles. Try Again. 3
Current pile size: 8 marbles. I'll take: 4
Current pile size: 4 marbles. How many would you like to take? 1
Current pile size: 3 marbles. I'll take: 1
Current pile size: 2 marbles. How many would you like to take? 1

Current pile size: 1 marbles. I'll take: 1

Current pile size: 0 marbles
YOU BEAT ME!!

Thank you for playing
```

Write a console-mode program, (so you don't need to get bogged down with graphics or the user interface; instead, you'll concentrate on designing the classes and the logic), in which the computer plays against a human opponent. Generate a random integer between 10 and 100 to denote the initial size of the pile. Generate a random integer between 0 and 1 to decide whether the human or the computer takes the first turn. Generate a random number between 0 and 1 to decide whether the computer plays *smart* or *stupid*.

In stupid mode, the computer simply takes a random legal value (between 1 and $n/2$) from the pile whenever it has a turn. In smart mode the computer takes off enough marbles to make the size of the pile a power of two minus 1 (that is, 3, 7, 15, 31, or 63). That is always a legal move, except if the size of the pile is currently one less than a power of 2. In that case, the computer makes a random legal move.

Note that the computer cannot be beaten in smart mode when it has the first move, unless the pile size happens to be 15, 31, or 63. Of course, a human player who has the first turn and knows the winning strategy can win against the computer.

When you implement your game, use the classes **Pile**, **Player** and **Game**. The Game class should contain your **main()** method that starts the game playing. A player can be either: a) stupid, b) smart or c) human (use an enumerated type for this). Human **Player** objects will prompt for input, while stupid and smart players will simply play according the appropriate strategy.

Package your program (including the source code) in a single executable jar file named **LastFirst_hw27.jar** and email it to **cs170f08@gmail.com**. Use only the exact phrase: **LastFirst_hw27** as the subject line on your email. Replace **LastFirst** with your actual last and first name. You can create an executable jar file from either BlueJ or from Eclipse. Watch the online lesson this week to see how. Make sure that your **main()** method starts the game playing.